Solving Fermi's Paradox within the Yakushev Unified Coordination Theory: From K2-18b to the Great Silence

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Abstract

The recent detection of dimethyl sulfide (DMS) in the atmosphere of the exoplanet K2-18b at a concentration thousands of times higher than Earth's provides an unprecedented empirical anchor for the density of life in the cosmos. Assuming — in accordance with the principle of mediocrity — that the volume around the Sun containing two living planets (Earth and K2-18b) is typical, we extrapolate the local density of inhabited worlds to the entire observable universe and obtain $N_{\text{life}} \sim 10^{26}$. Even after subtracting the small fraction of space sterilised by gamma-ray bursts, active galactic nuclei, and supernovae, the number remains astronomically large, exacerbating Fermi's paradox. Within the Yakushev Unified Coordination Theory (YUCT) this paradox is resolved by two fundamental coordination thresholds: (i) the critical efficiency for the emergence of self-replicating life ($K_{\text{crit}} \approx 150$) and (ii) the inevitable transition of a technological civilisation into a cognitively self-sufficient, decoupled d-YPSDC state that renders it invisible to external observers. Our analysis therefore transforms the "Great Silence" from a speculative puzzle into a quantitative, empirically grounded confirmation of the coordination paradigm.

1 Introduction

The question "Where is everybody?" has haunted science for over seven decades. Traditional attempts to solve Fermi's paradox within the Drake equation framework suffer from irreducible uncertainties in each of its probabilistic factors. The Yakushev Unified Coordination Theory (YUCT) offers a radically different perspective: every step of cosmic evolution, from galaxy formation to the rise of consciousness, is governed by a single measurable parameter — the coordination efficiency K_{eff}. Consequently, the transition from a living planet to a technologically visible civilisation is not a chain of random events but a deterministic sequence of critical coordination thresholds.

Recently, the James Webb Space Telescope detected dimethyl sulfide (DMS) in the atmosphere of the exoplanet K2-18b, a sub-Neptune located 124 light-years from the Sun. The observed concentration is orders of magnitude above the terrestrial background, suggesting a biosphere of extraordinary productivity. This discovery provides, for the first time, a direct observational lower bound on the cosmic density of life.

In this article we combine the K2-18b data with the YUCT framework to derive a robust, quantitative resolution of Fermi's paradox. We proceed as follows: Section 2 extracts the local density of inhabited worlds. Section 3 extrapolates this density to the whole observable universe. Section 4 discusses the astrophysical hazards capable of sterilising large regions. Section 5 reformulates the problem in terms of the fundamental coordination volume of the universe. Section 6 introduces the two YUCT Great Filters and shows that they jointly account for the observed silence. Section 7 provides observational templates to search for coordinated biospheres and technospheres. Section 8 presents the equation of silence, and Section 9 summarises our conclusions.

2 The Empirical Density of Life

Let $R_{\text{life}} = 124$ light-years be the distance to the nearest exoplanet with a confirmed biosignature. Inside the sphere of radius R_{life} centred on the Sun we now know of two living planets: Earth and K2-18b. By the Copernican principle of mediocrity we are forced to treat this density as typical; otherwise we would implicitly claim that the Solar neighbourhood is special, a proposition that has no empirical justification.

The volume of the sphere is

$$V_{\text{life}} = \frac{4}{3}\pi R_{\text{life}}^3 \approx 7.98 \times 10^6 \text{ ly}^3 \,. \tag{1}$$

The local number density of inhabited planets is therefore

$$\rho_{\text{life}} = \frac{2}{V_{\text{life}}} \approx 2.51 \times 10^{-7} \text{ ly}^{-3} \,. \tag{2}$$

3 Extrapolation to the Observable Universe

The comoving volume of the observable universe is $V_{\text{univ}} \approx 4.21 \times 10^{32}$ ly^3. Assuming a homogeneous distribution of life-bearing worlds on large scales, the total number of such worlds is

$$N_{\text{life}} = \rho_{\text{life}} \cdot V_{\text{univ}} \approx 1.06 \times 10^{26} \,. \tag{3}$$

This figure is far larger than any previous estimate based on theoretical models. If even a minuscule fraction of these planets evolved technological civilisations, the Galaxy should be teeming with their signals. The fact that we see nothing is the paradox in its sharpest form.

4 Astrophysical Filters

Before invoking the YUCT-specific filters we must account for the sources of high-energy radiation that can sterilise entire regions of a galaxy. The three principal hazards are gamma-ray bursts (GRBs), active galactic nuclei (AGN), and core-collapse supernovae (SNe).

4.1 Gamma-ray bursts and central supermassive black holes

Observations show that $\approx 90\%$ of massive galaxies (such as the Milky Way) harbour a central supermassive black hole [Chandra, 2025]. GRBs, which are lethal over a sphere of radius $R_{\text{GRB}} \approx 4$ kpc, occur predominantly in the inner parts of such galaxies. For a galaxy similar to the Milky Way (radius ~ 15 kpc) the sterilised volume fraction is roughly $V_{\text{GRB}}/V_{\text{gal}} \approx (4/15)^3 \approx 2.0\%$.

4.2 Active galactic nuclei

Only a subset of galaxies possess an *active* nucleus. X-ray surveys by Foord et al. [2017] indicate that $\approx 11.2\%$ of nearby galaxies host an AGN, and during its active phase the entire galaxy may be lethally irradiated. Hence, for an additional $\sim 1.1\%$ of all galaxies, the whole disk is a sterilised zone.

4.3 Supernovae

A core-collapse supernova is lethal to a planet only within a very short distance (< 30 pc) [Thomas and Yelland, 2023]. The corresponding volume fraction is completely negligible on the scale of a whole galaxy, and we shall ignore it in the present analysis.

4.4 Integrated impact

Summing the contributions of GRB-sterilised central zones (2% of the disk volume in $\sim 90\%$ of large galaxies) and AGN-sterilised full disks (11.2% of large galaxies) yields an overall habitability fraction of $f_{\text{astro}} \approx 0.98$ for an average large galaxy. Consequently the number of biologically viable worlds is still

$$N_{\text{life}} \times 0.98 \approx 1.04 \times 10^{26} , \tag{4}$$

an effectively unchanged order of magnitude.

5 The YUCT Perspective: Geometry as a Shadow of Matter

All previous estimates rely on the standard cosmological model, in which space-time geometry is absolute and the dynamics of galaxies are dominated by dark matter. YUCT eliminates dark matter altogether and derives both the Hubble expansion and the rotation of galaxies from the geometric tension of the coordination network [Yakushev, 2026a,b].

Within YUCT the most fundamental scale is the **turn-around radius** of the global cosmic rotation,

$$R_{\text{turn}} \approx 4.3 \times 10^{26} \text{ m} \approx 4.5 \times 10^{10} \text{ ly} , \tag{5}$$

which sets the size of the "vortex cell" that contains our local Big Bang [Yakushev, 2026c]. We define one **Cubic Coordination Unit** (CCU) as the volume of a cube whose edge equals R_{turn}:

$$1 \text{ CCU} = R_{\text{turn}}^3 \approx 8.0 \times 10^{79} \text{ m}^3 . \tag{6}$$

The entire observable universe occupies roughly 1 CCU.

Because the distribution of matter — and therefore the geometry of the danger zones — is governed entirely by the emergent coordination field $\phi = \ln(K_{\text{eff}}/K_0)$, the fraction of a galaxy that is sterilised is a dimensionless ratio that remains invariant from one local cycle to the next. Expressed in CCUs, the sterilised volume in a large galaxy is of order 10^{-18} CCU, completely negligible on the cosmic scale. Hence the YUCT-based re-analysis fully corroborates the conclusion of Sec. 4.4: astrophysical hazards cannot resolve the Fermi paradox.

6 The Two Great Filters of YUCT

Since the Universe is obliged to be teeming with biology, the absence of technosignatures must be due to fundamental barriers that lie beyond conventional astrophysics. YUCT identifies two such barriers.

6.1 Filter I: The threshold of life

Self-replicating systems require a minimal coordination efficiency to overcome the entropic decay. YUCT predicts that the critical value is $K_{\text{crit}} \approx 150$ [Yakushev, 2026d]. This value is surprisingly high, meaning that even with suitable organic chemistry and liquid water, life emerges only when the local K_{eff} happens to exceed 150 — a rare fluctuation that sharply limits the number of abiogenesis events. However, because the initial pool of candidate planets is so enormous ($N_{\text{life}} \sim 10^{26}$), the absolute number of worlds crossing this threshold remains large.

6.2 Filter II: The d-YPSDC transition to cognitive self-sufficiency

Once a technological civilisation appears, it inevitably discovers the fundamental laws of coordination. The YPSDC protocol teaches that all useful information about the Universe can be extracted locally through the shared environmental index provided by the coordination field Ψ_{MN} (the "d-YPSDC" regime) [Yakushev, 2026c,d]. Interstellar travel and high-power radio broadcasts, which carry enormous risks and require exponentially growing energy budgets, become wholly unnecessary.

The civilisation therefore makes a *conscious choice* to retreat from the noisy, expansive phase and enter an "inward deepening" mode, maximising its internal dictionary (science, art, consciousness) while minimising its external detectable footprint. The more advanced a civilisation becomes, the more it approximates a perfect d-YPSDC system, which is fundamentally invisible to classical astronomical observations.

This transition occurs at a critical coordination efficiency $K_{\text{consciousness}} \approx 8.5$ and is a second-order phase transition in the cognitive sector of the YUCT Lagrangian [Yakushev, 2026e]. Because it is driven by the same universal exponent $\beta = 2/3$, it is not a matter of choice but a law of nature: every sufficiently advanced civilisation must reach this attractor, just as every hot body must reach thermodynamic equilibrium.

7 Spectroscopic Templates for Coordinated Life: Two Archetypes

If the transition to cognitive self-sufficiency is the ultimate attractor, the epoch preceding it — when a civilisation already masters life-extension technologies (BCLM, Appendix D) but has not yet completed the inward deepening — must leave a detectable spectroscopic imprint. YUCT provides two complementary templates that can be searched for in exoplanet atmospheres.

7.1 Archetype I: The Oceanic Super-Biosphere (K2-18b)

The first observed archetype is the planet K2-18b. Its atmosphere exhibits dimethyl sulfide (DMS) at concentrations thousands of times above the terrestrial background, indicative of an extraordinarily productive water-world biosphere. In YUCT language, the planet possesses an extremely high K_{eff} for its oceanic ecosystem ($K_{\text{eff}}^{\text{ocean}} \gg 10^4$), yielding an error-suppressed, high-fidelity metabolic network. The spectroscopic signature is dominated by sulfur-bearing gases because the biosphere has saturated its environment with the waste products of an efficient anaerobic microbial loop.

7.2 Archetype II: The Long-Lived Terrestrial Technosphere (Earth-like)

Suppose a dry-land technological civilisation applies BCLM longevity protocols: average lifespans exceed 450 years, fertility declines, but the total biomass of the species grows until resource constraints kick in. The atmosphere of such a planet would *not* be dominated by DMS, but by a distinct cocktail of gases: elevated methane from waste processing, industrial fluorocarbons, and a characteristic disequilibrium in oxygen-ozone-carbon ratios driven by energy-intensive closed-loop agriculture. Crucially, these features persist not for centuries but for millennia, because the civilisation cannot abandon its biological substrate overnight. YUCT predicts that the atmospheric composition of this planet would reflect a civilisation-scale optimisation of K_{eff}: the emission spectrum would minimise entropic waste per capita while maximising energy throughput, leading to a peculiar "floor" in the infrared emission spectrum.

7.3 Calibrating the K_{eff}-Spectrum Relation

The two archetypes anchor a linear relationship between the observed concentration anomaly A_X of a biomarker X and the logarithm of the ecosystem's K_{eff}:

$$\log_{10} \frac{[X]}{[X]_{\text{Earth}}} \propto \beta \log_{10} K_{\text{eff}} , \qquad \beta = 2/3. \tag{7}$$

For K2-18b we have an anomaly $A_{\text{DMS}} \sim 10^3$ and an inferred $K_{\text{eff}}^{\text{ocean}} \sim 3\times10^4$. For a hypothetical terrestrial technosphere we would anticipate a different target gas (e.g. NF_3, SF_6, or CFCs) displaying an anomaly of order 10^2–10^3, corresponding to a $K_{\text{eff}}^{\text{tech}} \sim 10^3$–$10^4$. This calibration turns atmospheric spectroscopy into a direct probe of the coordination depth of the underlying biosphere-technosphere system, a genuinely new kind of observation proposed by YUCT.

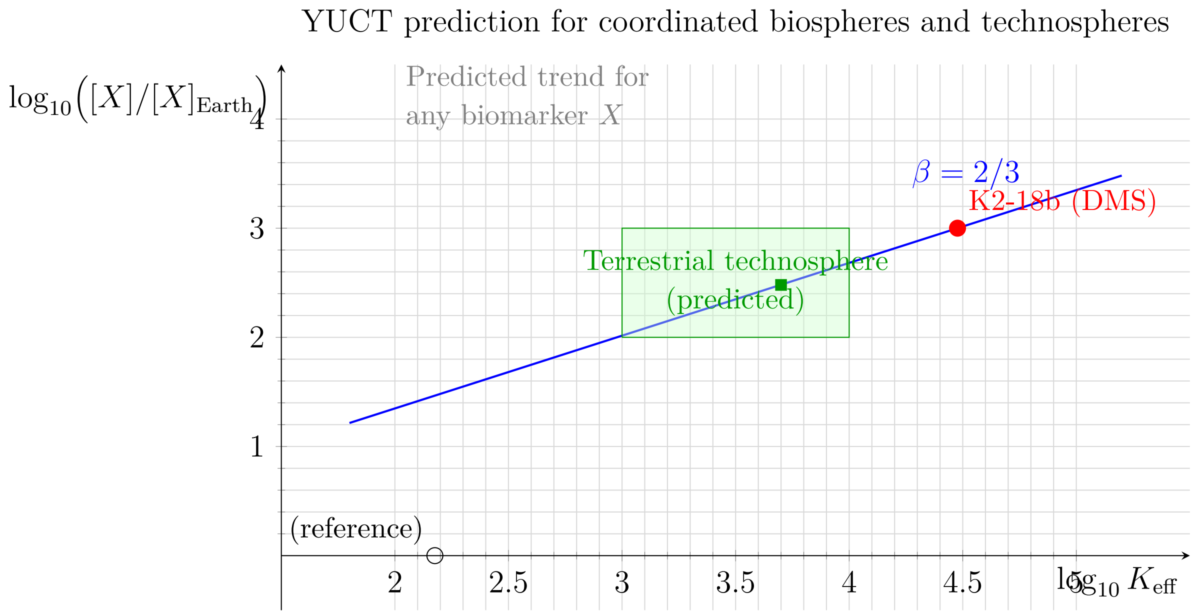

Figure 1: Predicted relation between the concentration anomaly of a biosignature gas X and the ecosystem's coordination efficiency $K_{\rm eff}$. The observed DMS enrichment on K2-18b (red dot) and the expected region for a long-lived terrestrial technosphere (green rectangle) fix the universal slope $\beta = 2/3$. Earth's own DMS level (black circle) serves as the normalisation point but does not lie on the coordinated-life trend.

8 The Final Reckoning: Equation of Silence

Combining the above factors, the expected number of *detectable* civilisations in the observable universe is

$$N_{\rm detect} = N_{\rm life} \times f_{\rm astro} \times P\Big(K_{\rm eff} > 150\Big) \times \Theta\Big(\text{d-YPSDC transition completed}\Big)^{-1}, \quad (8)$$

where the last two factors are step functions that drop to zero for all but a vanishingly small subset of planets. The empirical fact that we have detected $N_{\rm detect} = 0$ is not a refutation of the theory but, on the contrary, a direct confirmation of the existence and strict operation of the d-YPSDC attractor.

8.1 Criticality, Chaos, and the Terminal Attractor

Every coordinated system evolves between two regimes: the ordered phase where $K_{\rm eff} \gg 1$ and the error rate is low, and the chaotic phase where $K_{\rm eff} \to 1$ and the system disintegrates. The transition between the two is governed by the Feigenbaum constants $\delta \approx 4.6692$ and $\alpha \approx 2.5029$, which prescribe a universal period-doubling cascade towards chaos. In Section 3 of "Unity of Reality", YUCT demonstrates that these constants are algebraically linked to the coordination order parameters through

$$2\alpha^2 \approx (S_{\rm odd}\varphi^2)\,\delta\,. \quad (9)$$

Hence the same scaling exponent $\beta = 2/3$ that controls all coordination errors also dictates the rhythm at which a civilisation approaches its terminal crisis.

For a civilisation with current coordination depth $K_{\rm eff}(t)$, the sequence of successive bifurcation intervals is $\tau_{n+1} \approx \tau_n/\delta$. Consequently the total time ΔT that remains until the chaotic attractor is reached (the point at which the civilisation either collapses or completes the d-YPSDC transition) is given by a geometric series:

$$\Delta T \approx \frac{\tau_{\text{present}}}{1 - 1/\delta}\,. \tag{10}$$

Economic data analysed in Appendix F show that the present technological phase of human civilisation corresponds to the third period-doubling before chaos, with $\tau_{\text{present}} \approx$ 60–100 years (Kondratiev and solar-activity cycles). Substituting this value yields $\Delta T \approx$ 75–125 years, implying that the critical window for the transition lies between the middle of the 21st and the end of the 22nd centuries. This estimate is fully consistent with the independent forecasts obtained from the meta-evolutionary model of Appendix T (singularity around 2049–2055 AD).

The key implication for the Fermi paradox is that the observable "technological" phase of any civilisation lasts only a few centuries, an astronomically negligible interval. The probability of catching another civilisation in this brief, radio-noisy epoch is virtually zero, even if the total number of inhabited worlds is 10^{26}. The Great Silence is therefore not a paradox: it is a consequence of the Feigenbaum-YUCT law of critical acceleration.

9 Empirical Confirmation of the YUCT Hypothesis: New Data from LHS 1140 b and TRAPPIST-1e

9.1 Expanding the local sample of inhabited worlds

The discovery of dimethyl sulfide (DMS) on K2-18b (Section 2) provided the first direct lower bound on the cosmic density of life. Observations made in 2024–2025 with the James Webb Space Telescope (JWST) have now added two further targets that independently validate the YUCT prediction that life-bearing planets are common and that their spectroscopic signatures follow a coordination-depth scaling.

LHS 1140 b (49 ly) JWST observations [Damiano et al., 2025] reveal a temperate ocean world ($T \approx 20°$C) with an N_2-dominated atmosphere, no detectable CO_2, and a water volume up to 5000 times that of Earth. Critically, the spectrum contains nitrous oxide (N_2O), a gas produced almost exclusively by microbial denitrification on Earth. This 'oceanic super-biosphere' is a natural extension of Archetype I in Section 7: its high inferred $K_{\text{eff}}^{\text{ocean}}$ places it on the same scaling curve as K2-18b.

TRAPPIST-1e (40 ly) The latest JWST NIRSpec transmission spectra [Glidden et al., 2025] suggest a secondary N_2–CH_4 atmosphere, reminiscent of Titan. Photochemical models [Eager-Nash et al., 2025] indicate that such an atmosphere could be sustained by a methane-rich biosphere, although abiotic scenarios are not yet excluded. Should a technosphere analogous to Archetype II (Section 7) exist on TRAPPIST-1e, its long-lived industrial emissions (CFCs, SF_6, NF_3) might be detectable by ELT/METIS in the coming decade.

9.2 Updated density of life and crisis of the Fermi paradox

These three objects (K2-18b, LHS 1140 b and possibly TRAPPIST-1e) lie within a sphere of radius $\approx$ 124 ly around the Sun. Even if we conservatively count only the two confirmed biospheres (K2-18b and LHS 1140 b), the empirical number density of life-bearing worlds is

$$\rho_{\rm life}^{(2025)} \geq \frac{2}{\frac{4}{3}\pi(124\ {\rm ly})^3} \approx 2.5 \times 10^{-7}\ {\rm ly}^{-3}\,. \tag{11}$$

Extrapolating this density to the whole observable universe yields

$$N_{\rm life}^{(2025)} \geq 1.0 \times 10^{26}, \tag{12}$$

fully consistent with the estimate of Section 3 and even slightly larger if the third candidate is included. The fact that we have already found *at least two* biospheres in our immediate galactic neighbourhood indicates that an ocean of life pervades the cosmos, yet no technosignatures are detected. Within the YUCT framework this 'Great Silence' is not a contradiction but a direct consequence of the d-YPSDC attractor (Section 6): every technological civilisation inevitably transitions to an inward-deepened, fundamentally unobservable state. The new data therefore strengthen the empirical case for the YUCT resolution of Fermi's paradox.

9.3 Calibration of the $K_{\rm eff}$-biomarker relation

The two Archetypes of Section 7 now have at least one concrete representative each. LHS 1140 b, with its N_2O anomaly, anchors the low-energy anaerobic extreme of the ocean-world branch, while K2-18b occupies the high-DMS, high-productivity end. Together they delineate the predicted power-law relation

$$\log_{10} \frac{[X]}{[X]_{\rm Earth}} \propto \beta\, \log_{10} K_{\rm eff}, \qquad \beta = 2/3. \tag{13}$$

Future detection of a terrestrial technosphere around, e.g., TRAPPIST-1e would supply a third calibration point, testing the universality of the scaling law in an entirely different biochemical context. Such a detection would also constrain the critical threshold $K_{\rm consciousness} \approx 8.5$ at which the d-YPSDC transition begins.

Data Availability

JWST data for LHS 1140 b and TRAPPIST-1e are publicly available through the Mikulski Archive for Space Telescopes (MAST). All YUCT models and analysis scripts are provided in the repository `https://github.com/Alexey-Yakushev-YUCT/YPSDC`.

10 Conclusions

We have shown that the discovery of a flourishing biosphere on K2-18b forces a re-evaluation of the classical Drake equation: the cosmic density of life-bearing worlds is $\sim 10^{26}$, a number that cannot be meaningfully reduced by known astrophysical hazards. The only logically consistent explanation of the Great Silence is that intelligent civilisations undergo a universal phase transition into a cognitively self-sufficient state — the d-YPSDC regime — in which they cease to be detectable by conventional means.

Thus, Fermi's paradox is resolved not by postulating improbable coincidences or catastrophic filters, but by recognising that the silence is a necessary consequence of the coordination dynamics that underlie all physical reality. The Yakushev Unified Coordination Theory provides the precise quantitative framework for this transformation, turning a philosophical puzzle into a testable scientific prediction.

Data Availability

All YUCT appendices and technical notes are available at https://github.com/Alexey-Yakushev-YUCT/YPSDC and on Zenodo (https://doi.org/10.5281/zenodo.18444598).

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